ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN CYBER SECURITY

CIT3351 SOFTWARE AND SYSTEM SECURITY

FINAL EXAMINATION 2025

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 2 JUNE, 2025

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

QUESTION 1

Study the scenario below and answer the following questions:

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The average American household contains at least 10 connected devices, including laptops, phones, and smart devices. And every single one of those devices is an entryway for a cyber-attack. Cyber-attacks have not only become more prevalent, they have also become easier to pull off. Hackers can buy "subscription-based" malware attacks on the **Dark Web** as easy as signing up for a Netflix or Spotify account. According to a NETSCOUT Threat Intelligence report, there were 4.83 million cyber-attacks in the first half of 2022.

How do cyber-attacks happen?

Cyber-attacks occur when an individual, group, or organized gang attempts to maliciously breach the security system of another person or organization. While this might prompt you to envision hackers breaking into an online banking system to steal billions, the most common example of a cyber-attack is a data breach. Data breaches take place when hackers bypass a company or organization's security and steal sensitive information. They use this information for extortion, to commit other frauds, or to sell it on the Dark Web.

In 2021, the number of data breaches rose by a staggering 68%. But data breaches are only one of the consequences caused by cyber-attacks. Attacks can be used to gain personal information and allow cybercriminals to commit identity theft or cripple an organization's networks. Recently, there has been high prevalence of cyber warfare. This is when nation-backed hackers attempt to leak sensitive data, destroy computer networks, and even shut down banking and power infrastructure. The damage from these attacks can be severe. On average, data breaches cost companies over $4 million. Individuals, too, could potentially lose everything if a hacker gets access to their online accounts and sensitive information such as Social Security numbers and birthdays.

Below are some examples of cyber security breaches and attacks in America:

- In May 2021, JBS USA, the world's largest meat supplier, was hit with a ransomware attack that shut down production at many of its plants. The company ended up paying a ransom of $11 million in Bitcoin to prevent further damage.

- In one other extreme example in 2022, a hacker intercepted communications between a Chinese investor and a startup founder and got them to change the destination of a $1 million wire transfer.
- In the first few months of 2022, Microsoft, Google, and Apple all had to patch zero-day bugs. A zero-day attack occurs when hackers use known vulnerabilities to get into a system to steal data or cause damage.
- In 2016, a drive-by download attack used vulnerabilities in Adobe Flash Player to install crypto-ransomware. Once installed, victims were redirected to a site demanding 0.05 bitcoin to return access to their device.
- Lastly, in 2016, an Android device malware was discovered to be downloading rootkits to users' devices, leading to the theft of over one million Google account details.

Having read and understood the scenario above, answer the following questions:

a) From the scenario above, explain a cyber-attack. (4 marks)

b) Discuss the five (5) most common cyber security breaches in the world. (10 marks)

c) As stated above, in the first few months of 2022, Microsoft, Google, and Apple all had to patch zero-day bugs. A zero-day attack occurs when hackers use known vulnerabilities to get into a system to steal data or cause damage. Discuss some of the best three (3) techniques these organizations would have used to protect against zero-day attacks. (6 marks)

d) As stated above, in May 2021, JBS USA, the world's largest meat supplier, was hit with a ransomware attack that shut down production at many of its plants. The company ended up paying a ransom of $11 million in Bitcoin to prevent further damage. Explain five (5) consequences of a ransomware attack to a business organization such as this one. (10 marks)

e) There are five (5) features of computer security. In relation to the advent of cyber-attacks, discuss how these features, and if properly implemented could help in securing against the rampant cyber-attacks. (10 marks)

[Total: 40 marks]

[Total: 40 marks]

SECTION B: Answer any three (3) questions in this section.

QUESTION 2

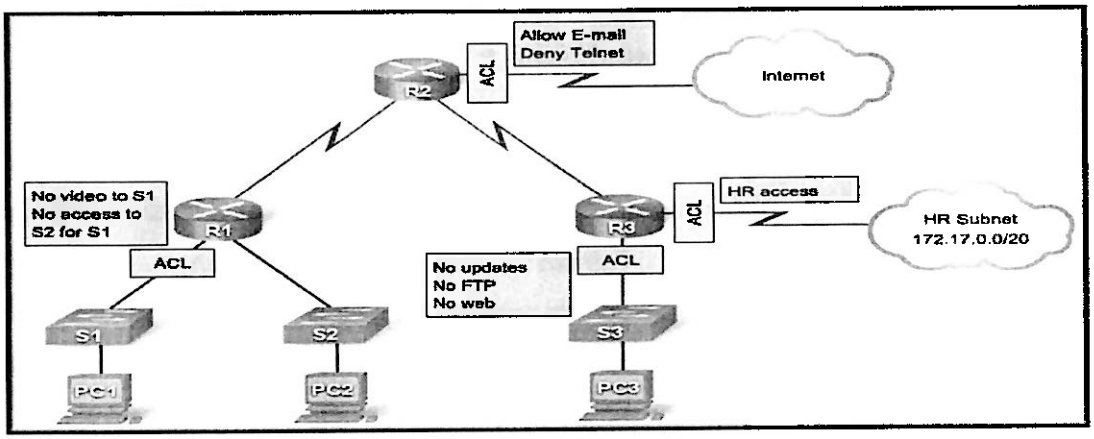
a) Just recently, your organization has been receiving reports that cyber criminals are using certain software to capture the managing director's business transactions on the computer and forwarding them to rogue device. Discuss this type of attack and how to prevent it. (6 marks)

b) A betting company has decided to purchase a firewall due to the advent of persistent cyber-attacks on its networks. Discuss the three (3) implementations of firewalls. (9 marks)

c) Microsoft is a certification authority (CA) for many organizations that use Asymmetric cryptosystems. Discuss any two (2) duties and importance of a Certificate Authority (CA) in crypto system implementation. (5 marks)

QUESTION 3

Access Control Lists (ACLs) enable you to control traffic into and out of your network. An ACL can be as simple as permitting or denying network hosts, IP addresses and controlling network traffic based on the TCP/UDP port being used.



a) According to the diagram above, write down an ACL that can do the following:

i) Allow traffic from all devices on network 172.17.0.0 and block all others (2 marks)

ii) Deny Hypertext Transfer Protocol (HTTP) traffic from network 10.0.0.0 (3 marks)

iii) Deny telnet traffic from network on R2, allow emails (3 marks)

iv) Deny FTP traffic from passing through R3 from HR (3 marks)

v) Deny simple mail transfer protocol traffic from network 172.17.0.0 (3 marks)

b) Access Control Lists (ACLs) limit network traffic and increase network performance based on the following three (3) parameters. Explain these parameters.

i) One ACL per protocol (2 marks)

ii) One ACL per Direction (2 marks)

iii) One ACL per Interface (2 marks)

QUESTION 4

a) Monitoring security and ensuring minimal company downtime is a huge challenge. Your organization is experiencing some unexplained new users on the wi-fi network. Discuss the measures you would put in place to prevent these unauthorized users. (6 marks)

b) Low-level security can be implemented using languages such as C, C++, Assembly, or Rust, or using libraries and frameworks that provide low-level functionality. Discuss the concept of low-level security. (5 marks)

c) Buffer overflow attacks are one of the low-level security vulnerabilities. With the use of one (1) example, describe tis attack. (4 marks)

d) According to research conducted by FBI, 70% of threats or attacks in organizations are internally motivated. With the use of three (3) examples, discuss the internal threats in computer security. (5 marks)

[Total: 20 marks]

QUESTION 5

a) Cryptography is the art and science of introducing secrecy in communication against third parties and adversaries. There are two (2) types of crypto systems, these are symmetric and asymmetric. With the use of two (2) examples of their application in industry, discuss these concepts. (8 marks)

b) In February 2020, Amazon Web Services (AWS) was the target of a large-scale distributed denial of service (DDoS) attack. The company experienced and mitigated a 2.3 Tbps (terabits per second) DDoS attack, which had a packet forwarding rate of 293.1 Mpps and a request rate per second (rps) of 694,201. It is considered one of the largest DDoS attacks in history.

i) Explain any two (2) types of DoS attack that might have been used in the scenario above. (4 marks)

ii) Discuss two (2) methods that could have been used to mitigate such attacks. (4 marks)

iii) Discuss any two (2) consequences that the attack had on Amazon services. (4 marks)

[Total: 20 marks]